

MarketGyan: Technical Experiment Progress Report

Compact sample weekly update: reproducible code, experiments, results, logs, and next actions

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Repository	News_aggregation	Branch	codex/market-ryan
Remote status	HEAD matches origin/codex/market-ryan		

Outcome. Delivered a bilingual, evidence-grounded NEPSE research prototype with a frozen 500-record benchmark, reproducible 350/75/75split, multilingual classifier baselines, a Qwen3.5-9B LoRA experiment, constrained inference, and a live single-pass RAG evaluation. Structure and citation gates passed; retrieval quality and neutral-direction performance remain below deployment thresholds. The correct status is **demo-ready research prototype**, not production financial advice.

Dataset gate	Python tests	Backend tests	Frontend tests
PASS	74 passed	98 passed	13 passed
500 valid records	0 failures	1 intentional skip	0 failures

Work completed and checked into GitHub

The reporting window focused on model reliability, experiment reproducibility, and an end-to-end demonstration path. All listed commits are present on `origin/codex/market-ryan`; this report references commit `4031ea7`.

Commit	Date	Delivered change	Research/engineering value
8ce52a5	1 Jul	Added the Qwen3.5-9B targeted-v2 A100 profile, strict metrics, model-gate logic, and training tests.	Reproducible GPU experiment configuration and comparable outputs.
e99f60e	7 Jul	Added final demo artifacts, proposal-evaluation code, vLLM deployment scripts, report evidence, and agent tests.	Connects model evaluation to an inspectable application workflow.
df2d1cc	7 Jul	Hardened filters, collectors, reconciliation, Qdrant retrieval, dashboard behavior, and XLM-R direction training.	Improves runtime observability and experiment iteration.
02796bc	7 Jul	Updated dataset handling and generated XLM-R notebook workflow.	Keeps notebook runs aligned with reusable Python modules.
4031ea7	8 Jul	Replaced label-name heuristics with count-driven, capped square-root oversampling; expanded unit coverage.	Makes class balancing data-driven and deterministic.

Implementation stack: Python, PyTorch/Transformers, XLM-R, FinBERT, Qwen3.5-9B, LoRA/Unsloth, vLLM, FastAPI, Node.js, React, MongoDB, Qdrant, Git/GitHub, and LaTeX.

Dataset and reproducibility gate

Only adjudicated schema-v2 records enter training. Candidate generation, reviewer annotations, adjudication, and export are separate states; duplicate groups remain intact across splits.

Check	Measured result	Acceptance rule / interpretation	State
Schema and evidence validation	500 valid; 0 issues	Required fields, ontology values, source metadata, symbols, and sentence IDs validate.	PASS
Relevance composition	300 direct / 100 indirect / 100 not relevant	Exact benchmark quota.	PASS
Language coverage	299 English / 200 Nepali / 1 mixed	At least 200 English and 200 Nepali records.	PASS
Symbol-level coverage	273 records	Minimum 150.	PASS
Source concentration	Largest source 295/500 (59%)	Maximum allowed share 60%.	PASS
Balanced grouped split	350/75/75	70/15/15 with duplicate-group isolation.	PASS
Neutral support	23 relevant records	Recommended floor 40; current support is too small for a reliable standalone class.	PARTIAL

Pipeline contract: Collect → normalize/provenance → generate candidate → independent review → adjudicate → validate/gate → grouped split → train/evaluate → retrieve/cite.

Experiment matrix and results

All model comparisons use the frozen 75-row test split unless noted. Strict JSON validity and evidence grounding are deployment gates; tolerant repair metrics are excluded from official results.

Condition	Validity	Grounding	Rel. F1	Dir. F1	Conclusion
XLM-R relevance classifier	–	–	0.743	–	Strong multilingual baseline; indirect class remains hardest.
XLM-R direction classifier	–	–	–	0.568	Three-class target merges neutral and uncertain because neutral support is sparse.
Qwen3.5-9B base, zero-shot	0.000	0.000	0.367	0.506	Classification signal exists, but output contract fails completely.
Qwen3.5-9B base, three-shot	0.987	0.987	0.435	0.502	Prompt examples fix format but not task quality.
Qwen3.5-9B targeted-v2 LoRA	1.000	1.000	0.796	0.575	Best balanced model-only result; structural gate passes.
LoRA via constrained vLLM	0.987	0.987	0.819	0.585	Schema safety retained; content metrics do not improve uniformly.

Table 1: Targeted-v2 Qwen3.5-9B model-only result: strict metrics.

Event macro-F1	Sector micro-F1	Symbol micro-F1	Evidence micro-F1	Nepali relevance acc.
0.633	0.677	0.793	0.736	0.788

Experiment conclusion. Fine-tuning is responsible for the large jump in relevance macro-F1 (0.367 zero-shot → 0.796 LoRA) and for strict output/grounding reaching 1.000. However, direction macro-F1 is only 0.575 and neutral recall is 0.000. Sector, symbol, and evidence extraction also show a recall gap. These failures block a deployment claim even though the structural gate passes.

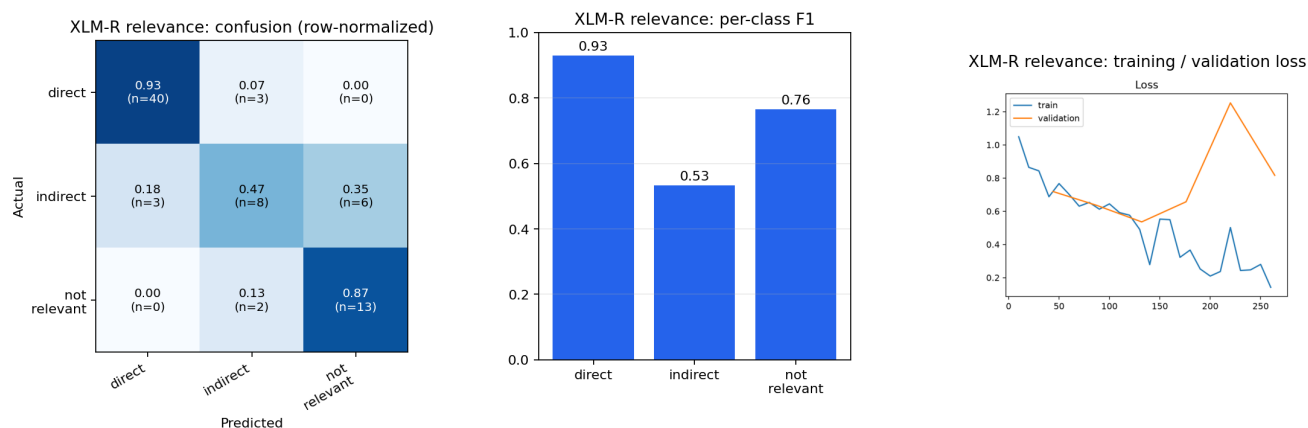


Figure 1: XLM-R relevance: normalized confusion matrix, per-class F1, and train/validation loss. The indirect class (F1 0.53) is the main classifier error source; late validation-loss growth indicates overfitting risk.

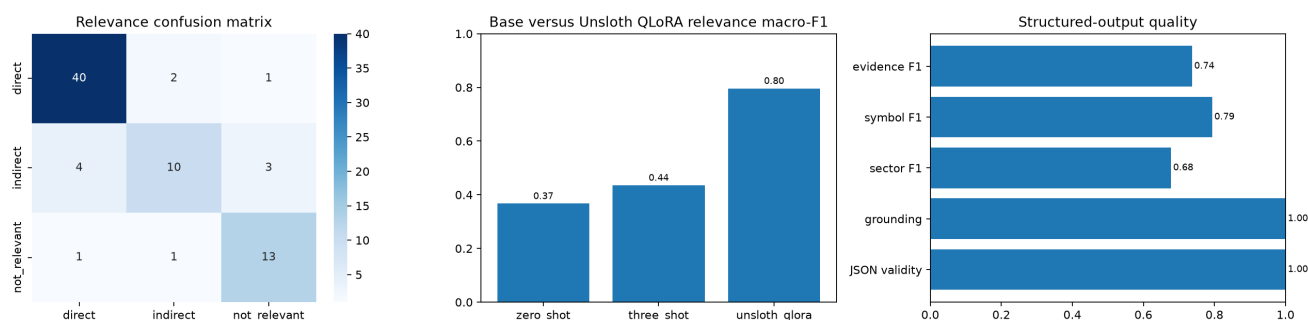


Figure 2: Qwen ablation and strict targeted-v2 test summary. LoRA materially improves relevance and structured extraction, but set-valued outputs remain below the project targets.

Live system experiment

The application experiment used a single-pass RAG agent: Qdrant returns bounded evidence rows, Qwen selects evidence indexes, and application code materializes exact citations. This prevents the model from inventing URLs or sentence identifiers.

Measure	Observed result	Interpretation	State
Scenario suite	20/20 passed	Schema, citation, sentence-citation, grounding, disclaimer, and no-advice checks all reached 1.000.	PASS
Latency	6.67 s average; 9.74 s p95	Acceptable for a research demo; requires production profiling and capacity testing.	PARTIAL
Judged-query retrieval P@5	0.262 across 13 query groups	Below the 0.60 proposal target. Metadata/filter coverage is the primary bottleneck.	BLOCKED
All-query zero-padded P@5	0.113 across 30 queries	Penalizes unjudged/empty groups and confirms retrieval is not deployment-ready.	BLOCKED

Execution log

The following condensed log was produced from the current checkout. Live provider calls and GPU training were not rerun; their retained JSON metrics and report plots are treated as experiment artifacts.

```
$ PYTHONPATH=. python3 -m market_gyan.cli validate data/processed/nepse-impact-500.jsonl
valid=true | records=500 | relevance=300/100/100 | language=en:299,ne:200,mixed:1

$ PYTHONPATH=. python3 -m market_gyan.cli gate data/processed/nepse-impact-500.jsonl --min-records=500
ready=true | errors=0 | warning="neutral relevant records 23 < recommended floor 40"

$ PYTHONPATH=. python3 -m unittest discover -s tests
Ran 74 tests in 0.353s | OK

$ npm test -- --test-reporter=spec # Node backend
tests=99 | pass=98 | fail=0 | skipped=1 (live Gemma credential-gated test)

$ CI=true npm test -- --runInBand --watchAll=false src/components/MarketGyanDashboard.test.js
test_suites=1 passed | tests=13 passed | failures=0
```

Recorded experiment artifacts:

- ml/marketGyan/outputs/proposal_eval/model_only_benchmark.json
- ml/marketGyan/outputs/proposal_eval/constrained_inference_metrics.json
- ml/marketGyan/outputs/proposal_eval/proposal-system-metrics.live_demo_report.json
- ml/marketGyan/data/processed/splits/manifest.json

Findings, issues, and next actions

Key findings

- Data-centric controls matter: a frozen, grouped split makes prompt and fine-tuning ablations comparable.
- Few-shot prompting solves much of the JSON-format problem but does not learn the Nepal-specific task.
- LoRA improves relevance and evidence extraction, yet rare labels and multi-value recall remain limiting.
- Citation materialization in application code is safer than trusting model-generated source metadata.
- Retrieval quality, not JSON formatting, is now the largest end-to-end blocker.

Next experiment cycle

1. Add and independently adjudicate neutral, indirect, regulation, and sector-industry hard cases; report agreement.
2. Run controlled XLM-R direction ablations for count-driven oversampling, merged labels, and class-weighted loss.
3. Build retrieval hard negatives, normalize metadata filters, and report per-query P@5/Recall@5 slices.
4. Tune set-valued selection thresholds for sectors, symbols, and evidence IDs without relaxing grounding rules.
5. Add CI publication of metrics, plots, test logs, and a one-page ClickUp-ready status summary.

Decision and communication. Continue the research prototype and demonstration work. Do not describe the system as deployable until retrieval reaches the agreed gate, neutral-direction support is expanded, and content metrics are revalidated on an unchanged held-out test split. No trading or investment action is performed by the system.